

Science Olympiad — Day 4 HW

Names of participants: (Please print neatly)

School Name:

Team Number:

Check your team: ☐ Varsity ☐ JV1 ☐ JV2 ☐ JV3

Warning: Do not open this packet until given permission to do so.

Note: There are useful notes after this page.

Question	Value	Incorrect letters	Deduction	Score
1	0			
2	0			
3	0			
4	0			
5	0		≡	
6	0		≡	
Final Score				

1) **[0 points]** You find a coded sequence scratched into the surface of a lab stool. It turns out to be a quote, written in a Baconian. What does it say?

ABBBABAABABABAABBABBABBABBABBBBABAABBABBABBBAABAABBAA

BAABBAABABBBABBAAABBBAAABBABBBAAABBBBBABBAAABBABBABBABB

AABABABABBAABAABBBABBABB

2) **[0 points]** While debugging a program, you find this hidden in the comments. It's a quote, encrypted with a Baconian. What does it say?

BaKëRBàKeRBàKëRBaķëR**B**àKeRBàKe**f**Baķë**f**BàķeRBàķëRBaķë**f**BaKe**f**

BàķëR**B**àKeR**B**àKëR**B**àķeR**B**àķëRBaKëR**B**àKeRBàķe**f**Baķë**f**BàKe**f**BàKeR

BàKe**f**BàKëRBàKëR**B**àķëRBaKëRBaķëRBàķeR

3) [0 points] Found on a microchip inside a disassembled robotics kit: this Baconian Cipher. What does it say?

Ä.Ö·ääi:·Äé°cäi:·ÄiÖä.i:··ÄÖääi:é:°ÄÖäci:·ÄiÖä.i:··Ä

Öäéäi°:·cÄÖäi:ı:·Ä·Öäiä:·Äé°äci:·ıÄÖäi.:·Ä·äéÖäi°:·Ä

Öäi:c·ÄÖäiı:··Ä·Öäi:·äÄÖéäi:·°ÄÖcäiı.:··äéÄÖä°i:·ÄÖci

äi:··äé:·Ä°Ö

4) [0 points] The janitor found this strange paper strip under a Bunsen burner. It's a quote, encoded with a Baconian Cipher. You are told that the deciphered text starts with I'm. What does it say?

AGONY AWFUL I FEEL BOOK A MARCO GNOME ANGRY AN OUT

MITCH COINS ALIAS ADOPT GREET ROHAN GREAT WRECK PHONY

WRATH FEWER QUITE DATED NIGEL RISKY FUNKY DO SAY

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Replacement																										

5) **[0 points]** This string of symbols appears on the display of a malfunctioning calculator. It hides a quote, encoded as a Cryptarithm. What do the values **68092 75413** decode to?

Values to decode for solution

6	8	0	9	2	7	5	4	1	3

Cryptarithm formula

$$\begin{array}{r}
 \text{M O T H E R} \\
 + \quad \text{M A L E} \\
 + \quad \text{M A L E} \\
 \hline
 \text{O R P H A N}
 \end{array}$$

	0	1	2	3	4	5	6	7	8	9
A										
E										
H										
L										
M										
N										
O										
P										
R										
T										

6) **[0 points]** The following quote needs to be decoded with the Cryptarithm Cipher. What do the values **465 9023 871** decode to?

Values to decode for solution

4	6	5	9	0	2	3	8	7	1

Cryptarithm formula

$$\begin{array}{r}
 \text{D I N G H Y} \\
 - \quad \text{K A Y A K} \\
 \hline
 \text{S H I P}
 \end{array}$$

	0	1	2	3	4	5	6	7	8	9
A										
D										
G										
H										
I										
K										
N										
P										
S										
Y										